2025 5th IEEE International Conference on Applied Electromagnetics, Signal Processing, & Communication (AESPC) | 979-8-3315-6719-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/AESPC67542.2025.11326839

On the Implementation of de Bruijn Sequence Generators with Higher Linear Complexity

Suman Roy
School of Electronics Engineering
KIIT Deemed to be University
Bhubaneswar, India
suman.royfet@kiit.ac.in

Arighna Deb
School of Electronics Engineering
KIIT Deemed to be University
Bhubaneswar, India
airghna.debfet@kiit.ac.in

Ujjwol Barman
Electronics & Communication Engineering
Tezpur University
Assam, India
ujjwol@tezu.ernet.in

Pradipta Dutta
School of Electronics Engineering
KIIT Deemed to be University
Bhubaneswar, India
pduttafet@kiit.ac.in

Trupti Mayee Behera
School of Electronics Engineering
KIIT Deemed to be University
Bhubaneswar, India
truptifet@kiit.ac.in

Poulomee Ghosh
Electrical Engineering
Indian Institute of Technology Delhi
New Delhi, India
Poulomee.Ghosh@ee.iitd.ac.in

***Abstract*—This paper presents a method for generating de Bruijn sequences with enhanced linear complexity using a word-based primitive σ-LFSR integrated with a nonlinear substitution box (S-box). Conventional σ-LFSRs, though efficient for hardware implementation, exhibit limited linear complexity, making them susceptible to cryptanalytic attacks. The proposed approach extends traditional LFSR-based de Bruijn sequence generation to a vectorized multi-input multi-output (MIMO) framework capable of exploiting the parallelism of modern processors. By inserting an all-zero vector into the state cycle of a primitive σ-LFSR and applying nonlinear bijective S-box transformations, the resulting vectorized de Bruijn sequence generators (DBSGs) achieve significantly higher linear complexities. Simulations validate the effectiveness of the proposed scheme, demonstrating that the component sequences are no longer cyclic shifts of one another, thereby increasing the overall linear complexity. The method offers a simple yet efficient architecture for generating high-complexity pseudorandom sequences suitable for cryptographic and communication applications.**

***Index Terms*—De Bruijn sequence, σ-LFSR, S-box, pseudo-random number generator (PRNG), linear complexity, stream cipher, cryptography.**

I. Introduction

Pseudorandom number generators (PRNGs) commonly employ linear feedback shift registers (LFSRs) as their fundamental building blocks. Sequences generated by primitive LFSRs, those governed by primitive characteristic polynomials, exhibit desirable statistical properties associated with randomness, including long periods, uniform output distribution, run and span-n properties, and a two-level autocorrelation function [1].

To exploit the parallel processing capabilities of modern architectures, Zeng et al. introduced a word-oriented stream cipher based on LFSRs incorporating multi-input multi-output (MIMO) delay blocks, referred to as the σ-LFSR [2]. Although similar to conventional LFSRs, the sequences generated by word-oriented variants generally possess low linear complexity (a measure of randomness), making them susceptible to cryptanalytic algebraic attacks capable of revealing their internal states [3]. Nevertheless, owing to their simplicity and efficiency in hardware realization, σ-LFSRs have been widely adopted in several stream cipher designs, such as SNOW 3G and Sosemanuk, typically in combination with nonlinear components to enhance security [4], [5], [6], [7].

In [8], a nonlinear substitution box (S-box)-based approach was proposed to mitigate the linearity inherent in σ-LFSRs and thus increase their linear complexity. Building upon this concept, the present work extends the S-box framework to develop a vectorized sequence generator. It has been empirically observed that these sequences exhibit a higher linear complexity. Therefore, such sequences have significant applications in keystream generation, exhaustive testing, randomness evaluation, side-channel resistance, and optimized input coverage in cryptographic systems. The proposed algorithm has been implemented and validated through software simulation in MATLAB & Simulink. Furthermore, the statistical properties and linear complexities of the generated sequences are examined to demonstrate their effectiveness in construction of attack-resistant pseudorandom number generators (PRNGs) suitable for secure cryptographic applications.

II. Methodology

This section presents the implementation methodology for generating de Bruijn sequences by employing a word-based primitive σ-LFSR defined over arbitrary finite fields, integrated with a nonlinear substitution box (S-box). The inclusion of the nonlinear transformation enhances the statistical properties of the output and leads to a noticeable increase in linear complexity of the generated sequences.

A. Generating de Bruijn Sequences

A de Bruijn, $DB_k(n)$, sequence of order n over an alphabet of size k is a cyclic sequence of period k^n in which each k-ary n-tuple occurs exactly once. De Bruijn sequences constitute a class of pseudorandom sequences known for their desirable

979-8-3315-6719-4/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Lakehead University. Downloaded on September 26,2026 at 19:46:26 UTC from IEEE Xplore. Restrictions apply.

statistical properties and relatively high linear complexity [9], [10]. Such sequences can be efficiently generated using LFSRs. Consider a conventional single-input single-output (SISO) primitive LFSR comprising L delay elements over the finite field $\mathbb{F}_2$. Let the linear feedback function be represented as $f(s_0, s_1, \ldots, s_{j+L-1})$. To construct a de Bruijn sequence, a feedback shift register (FSR) can be defined with the following feedback function h [1]:

$$h(s_0, s_1, \ldots, s_{j+L-1}) = f \oplus g(s_0, s_1, \ldots, s_{j+L-1}) \quad (1)$$

where $\oplus$ denotes addition over $\mathbb{F}_2$, and the nonlinear component g is given by

$$g = \overline{s_1 \oplus s_2 \oplus \ldots \oplus s_{j+L-1}}. \quad (2)$$

The nonlinear logic function g appends a zero to the subsequence of $(L-1)$ consecutive zeros present in the sequence of length $2^L - 1$ generated by the primitive LFSR (commonly referred to as the *m-sequence*). This modification transforms the m-sequence into a de Bruijn sequence possessing a period of 2^L. These maximal-length sequences possess a linear complexity of $LC + 1$, where LC denotes the linear complexity of the m-sequence generated by the underlying LFSR. The aforementioned method can be easily extended to a σ-LFSR configuration for generating vectorized de Bruijn sequences with enhanced linear complexity. The detailed formulation and implementation of this approach are presented in the subsequent sections.

B. σ-LFSR

A typical σ-LFSR configuration comprising L delay blocks, each with r-input and r-output connections, is shown below.

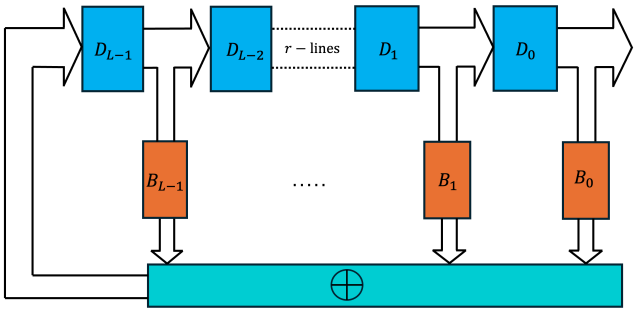


Fig. 1. An L-stage σ-LFSR with r-input, r-output delay blocks.

Here, the r output sequences over $\mathbb{F}_2$ are referred to as the *component sequences*. For a primitive σ-LFSR, these component sequences are cyclically shifted versions of one another [11], [12]. The linear complexity of the vectorized output sequence is equal to the least common multiple (LCM) of the linear complexities of its component sequences. Since all component sequences are generated using the same characteristic polynomial, the overall linear complexity of the σ-LFSR output remains low, typically equal to rL.

In [8], the authors proposed the use of a nonlinear substitution box (S-box) to enhance the linear complexity. In the following section, it is demonstrated that converting the σ-LFSR into a MIMO de Bruijn sequence generator and subsequently applying the S-box further improves the linear complexity of the generated sequence.

C. Implementing word-oriented de Bruijn Sequence Generator

As in the case of the conventional SISO LFSR discussed in the previous subsection, a similar approach can be employed to insert a zero vector into the vectorized m-sequence for generating a vector de Bruijn sequence. A schematic representation of this concept is shown in the figure below.

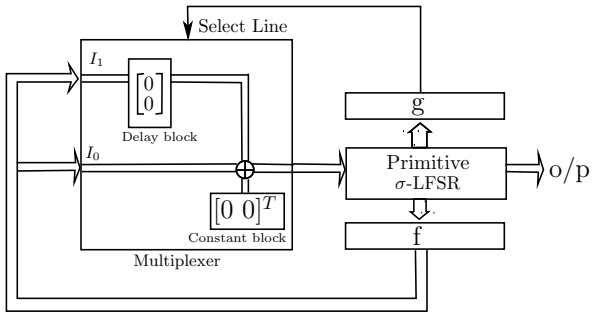


Fig. 2. A word-oriented de Bruijn sequence generator

A primitive σ-LFSR over $\mathbb{F}_2$ with a linear feedback function f is considered here. The output of f, which depends on the initial state and the feedback gain matrices of the σ-LFSR, is connected to both the inputs of a 2:1 multiplexer. Let the initial state vector of the primitive σ-LFSR be $\mathbf{s}(0) = [\mathbf{s}_0\ \mathbf{s}_1, \ldots\ \mathbf{s}_{L-1}]^T \in \mathbb{F}_2^{rL}$ where r is the number of inputs of each delay block and $\mathbf{s}_i = [s_{i0}, s_{i1}, \ldots, s_{i(r-1)}] \in \mathbb{F}_2^r$. Here $s_{ij} \in \mathbb{F}_2$ is the j-th output bit of the i-th multi-input multi-output delay block. Note that, out of multiple possibilities, an all-zero state vector can be inserted after the state vector $\widehat{\mathbf{s}(k)} = [\mathbf{s}_0(k) \underbrace{\mathbf{0}\ \mathbf{0}\ \ldots\ \mathbf{0}}_{L-1\ copies}]^T$ where $\mathbf{0} = [0\ 0\ \ldots 0]^T$ and $\widehat{\mathbf{s}_0(k)}$ is a nonzero vector in $\mathbb{F}_2^r$. Without loss of generality, let $\widehat{\mathbf{s}_0(k)} = [\underbrace{0\ 0\ \ldots\ 0}_{r-1\ copies}\ 1]^T$. With this assumption, one can obtain the nonlinear function g as follows.

$$g(s_{00}, \ldots s_{0(r-1)}, s_{10}, \ldots, s_{(L-1)(r-1)}) = \overline{\bigoplus_{m,n} s_{mn}}$$

where $m \in \{0, 1, \ldots, L-1\}$ and $n \in \{0, 1, \ldots, r-1\}$. Clearly, the nonlinear function g can be realised by a rL-input NOR gate, as the previous. Output of g selects the input channel for the multiplexer as shown in the Figure 2. Input I_1 is connected to a delay block that initialised with $\mathbf{0} \in \mathbb{F}_2^r$. The output of g will be 1 (high), for the state vector either $\mathbf{0}$ or $\widehat{\mathbf{s}(k)}$. When g detects $\widehat{\mathbf{s}(k)}$, $\mathbf{0}$ is bit-wise XORed with a constant zero vector and then transferred to the σ-LFSR. At the same time, the output of the linear function f is stored in the delay block. And, when g detects the all-zero state vector,

Authorized licensed use limited to: Lakehead University. Downloaded on September 26,2026 at 19:46:26 UTC from IEEE Xplore. Restrictions apply.

then the content of the delay block released to the σ-LFSR. For all other instances, g gives 0 (low) output. It selects the I_0 input that essentially acts as a regular feedback path of the σ-LFSR. Thus, a de Bruijn sequence, $DB_{2^r}(L)$, over $\mathbb{F}_q$ can be generated from a primitive σ-LFSR by adding an all-zero state vector into its state cycle. It is a de Bruijn sequence generator (DBSG).

Clearly, from the proposed scheme, the component sequences generated by DBSGs are no longer simple cyclic shifts of one another. Consequently, the overall linear complexity increases up to $\prod_{i=1}^{r} LC_i$, where LC_i denotes the linear complexity of the i-th component sequence. This observation is validated through software simulations in the following section.

III. Result and Discussion

Consider the primitive characteristic polynomial $\chi(s) = s^4 + s + 1$. We can design a σ-LFSR of length 2 having two 2-input, 2-output delay blocks with the following feedback matrices:

$$B_0 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We have implemented the above system in Simulink (Fig. 3).

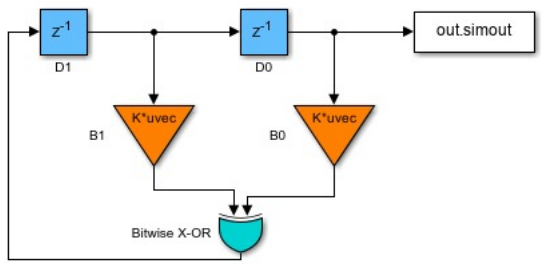


Fig. 3. A word-oriented de Bruijn sequence generator

The generated output sequence is as follows:

$$\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix}$$

Clearly, the proposed scheme will generate the following sequence.

$$\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}1\\1\end{bmatrix}$$

Upon applying the binary r-tuple to decimal mapping, we obtain a de Bruijn sequence of order $L = 2$ over $GF(4)$, namely $DB_4(2) = 1002212033230113$. The Berlekamp–Massey algorithm is then used to compute the linear complexities of the corresponding binary component sequences [13]. Table I compares the obtained results with those reported in [8], demonstrating a significant improvement in linear complexity. Furthermore, as discussed in [8], the application of nonlinear bijective S-boxes in conjunction with DBSGs further enhances the overall complexity of the sequence.

IV. Conclusion and Future Work

In this work, a novel approach for generating de Bruijn sequences using a word-based primitive σ-LFSR in conjunction with a nonlinear substitution box (S-box) has been presented. The proposed method effectively extends the conventional LFSR-based de Bruijn sequence generation technique to a vectorized framework, enabling efficient implementation on word-oriented processors. The simulation results confirm a substantial increase in the overall linear complexity. Furthermore, integrating nonlinear bijective substitution boxes enhances the cryptographic strength of the generated sequences.

The proposed architecture thus offers a promising solution for generating high-complexity pseudo-random sequences suitable for cryptographic and communication applications. Future work may explore hardware implementation aspects and optimization for specific cryptographic protocols. Also, it will be interesting to test the obtained sequences using the NIST statistical test suite to strengthen the randomness evaluation [14].

References

[1] S.W. Golomb, Shift register sequences, San Francisco:Holden-Day, 1967.
[2] G. Zeng, W. Han and K. He, High efficiency feedback shift register : σ-LFSR, IACR Eprint archive, 2007.
[3] C. Paar and J. Pelzl, Understanding Cryptography: A Textbook for Students and Practitioners. Berlin, Germany: Springer, 2010.
[4] J. K. Lee, D. Lee, and S. Park, Cryptanalysis of Sosemanuk and SNOW 2.0 Using Linear Masks, in Asiacrypt 2008, Lecture Notes in Computer Science, vol. 5350, pp. 30–49, 2008.
[5] M. Bahadori, K. Järvinen and V. Niemi, FPGA implementations of 256-bit SNOW stream ciphers for postquantum mobile security, in IEEE Trans. on Very Large Scale Integration (VLSI) Systems, vol. 29, no. 11, pp. 1943-1954, November 2021, doi: 10.1109/TVLSI.2021.3108430.
[6] C. Berbain, O. Billet, A. Canteaut, N. Courtois, H. Gilbert, L. Goubin, et al., Sosemanuk, a fast software-oriented stream cipher, In New Stream Cipher Designs. Springer; 2008. pp. 98-118.
[7] S. Nandi, S. Krishnaswamy, B. Zolfaghari, and P. Mitra, "The application of σ-LFSR in Key-Dependent Feedback Configuration for Word-Oriented Stream Ciphers," arXiv preprint arXiv:2003.09381, 2020.
[8] S. Pal, P. Ghosh, S. Roy, and P. Biswal, On the analysis of sequences generated by S-boxes cascaded with σ-LFSRs, in 2021 2nd International Conference on Applied Electromagnetics, Signal Processing and Communication (AESPC), Bhubaneswar, India, Nov. 2021, pp. 472–476.
[9] E. Key, An analysis of the structure and complexity of nonlinear binary sequence generators, IEEE Trans. on Information Theory, Vol. 22, pp. 732-736, November 1976.
[10] K. Mandal, B. Yang, G. Gong and M. Aagaard, Analysis and efficient implementations of a class of composited de Bruijn sequences, in IEEE Trans. on Computers, vol. 69, no. 12, pp. 1835-1848, December 2020, doi: 10.1109/TC.2020.2979460.
[11] S. Krishnaswamy and H. Pillai, On the number of linear feedback shift registers with a special structure. IEEE Trans. on Inf. Theory, vol. 58(3), pp. 1783-1790, 2012.
[12] S. Roy and S. Krishnaswamy, On the frequency of symbols in sequences generated by nonlinear feedforward generators, Cryptography and Communications, Springer, vol. 12(1), pp. 115-126, January 2020.
[13] A. J. Menezes, P. C. van Oorschot, and S. A. Vanstone, Handbook of Applied Cryptography. Boca Raton, FL, USA: CRC Press, 1996.
[14] Andrew Rukhin et al., A statistical test suite for random and pseudorandom number generators for cryptographic applications, Special Publication 800-22 Revision 1a., Computer Security, U.S. Department of Commerce, 2010.

Authorized licensed use limited to: Lakehead University. Downloaded on September 26,2026 at 19:46:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
FEEDBACK MATRICES AND LINEAR COMPLEXITIES FOR VARIOUS CONFIGURATIONS

Sl. No.	r	L	Char. poly., $\chi(s)$	B_0	B_1	B_2	σ-LFSR	σ-LFSR with S-box	DBSG	DBSG with S-box
1	2	2	$s^4+s+1=0$	$\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}$	$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$	—	4	4	30	99
2	2	3	$s^6+s+1=0$	$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$	$\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}$	$\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$	6	6	48	824
3	3	2	$s^6+s+1=0$	$\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$	$\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 1 & 0 & 0\end{bmatrix}$	—	6	21	24	2581
4	3	3	$s^9+s+1=0$	$\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$	$\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 1 & 0 & 0\end{bmatrix}$	$\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$	9	45	99	6168

Authorized licensed use limited to: Lakehead University. Downloaded on September 26,2026 at 19:46:26 UTC from IEEE Xplore. Restrictions apply.